

Portrayals of Race and Gender: Sentiment in 100 Years of Children's Literature by

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Research Question

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How have portrayals of race and gender in children's literature changed over the past 100 years, and how are these portrayals related to contemporary social beliefs about race and inequality?

Important to know

- **«Early exposure to stimuli shape children's beliefs about the world»**
- Different content in textbooks → different beliefs in adulthood
- Dominant groups in USA: white people and males
 - Children believe white males occupy dominant social roles

Important to know

- Skewed **representation** of social categories → prominent
 - Inequality in personal
 - Societal outcomes: labour force participation, mortality and incarceration

Important to know

Race and gender treated as **mutually exclusive** → Mistake



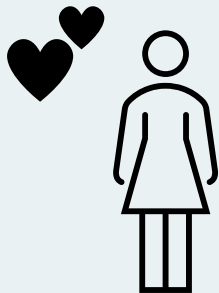
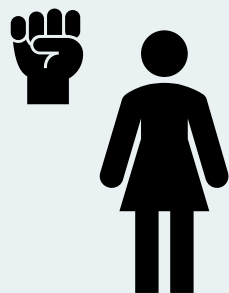
Intersectionality of identities



Individuals with multiple marginalized or excluded identities → **excluded**
from theoretical, analytical, and practical consideration

Important to know

- Children process identity categories and intersectionality **subconsciously**, they have **no focused cognitive** skills
 - Brain automatically links impressions and feelings to these identities.
 - heuristics and implicit biases



Books

- Recognition from the **American Library Association**
- Divide in two collections:
 - «**Mainstream**» books: Newbery Medal or Caldecott Medal
 - «**Diversity**» books: focus on underrepresented identities



Data

- Two levels:
 - **Collection** level → overall representation
 - **Collection-by-decade** level → changes in sentiment over time

Methods

- **Google Vision Optical Character Recognition** → detect and extract text
- **Pre-trained Punkt tokenizer** → split corpus into sentences
- **Famous figure extraction**
- Add **[CLS]** (classification) and **[SEP]** (Separator) tokens to start and end of sentences

Famous Figure Extraction

- What are the sentiments associated with different identities?
- **SpaCy's Named Entity recognition** (NER) pipeline to extract all «Person» entities
- Compare with publicly available data set **Pantheon 2.0**
 - 70'000 Wikipedia biographies
 - Fuzzy matching algorithm → similarity $\geq 90\%$

Famous Figure Extraction

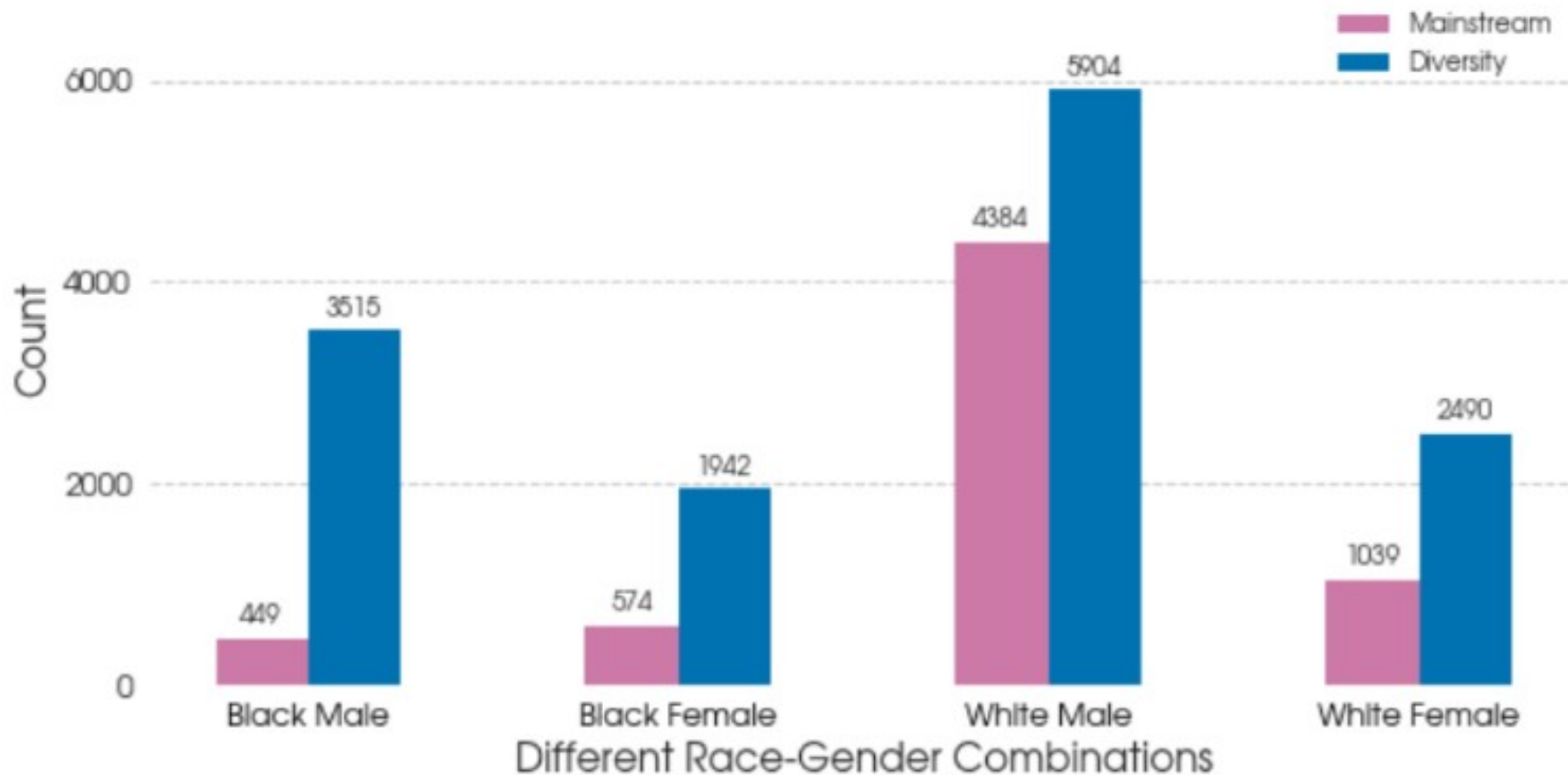
- Pantheon 2.0 doesn't contain **race**
- Manually code race for each person
- **Mainstream: 24'365 unique & 272'776 total Person counts**
 - **Famous: 925 unique & 5'526 total counts**
- **Diversity: 45,892 unique & 379,074 total Person counts**
 - **Famous: 2'267 unique & 11'061 total counts**

Sentiment Analysis Model Selection

- How are different identities portrayed?
- **Analysing sentiment** → race & gender
- **BERT: Bidirectional Encoder Representations from Transformers.**
 - Context around the word
 - trained to “understand” language itself by reading huge amounts of text

Fine-tuning BERT

- Not specifically made for sentiment analysis
- **Stanford Sentiment Treebank (SST-5)**
 - Movie reviews



(a) Number of Mentions, by Race-Gender



Table 1: Model Performance

Model	Accuracy	F-Score
bert-base-uncased	0.47	0.46
best-bert-model	0.43	0.41
bert-base-uncased-pytorch-trainer	0.45	0.42
bert-base-uncased-new-layers	0.46	0.44
roberta-base	0.39	0.37
vader	0.38	0.28
textblob	0.42	0.37
nlptown	0.26	0.30

Accuracy

- The percentage of all predictions that the model got **completely right**
- Accuracy alone can be **misleading** if one class dominates (if most sentences are neutral).

$$\text{Accuracy} = \frac{\text{Number of correct predictions}}{\text{Total number of predictions}}$$

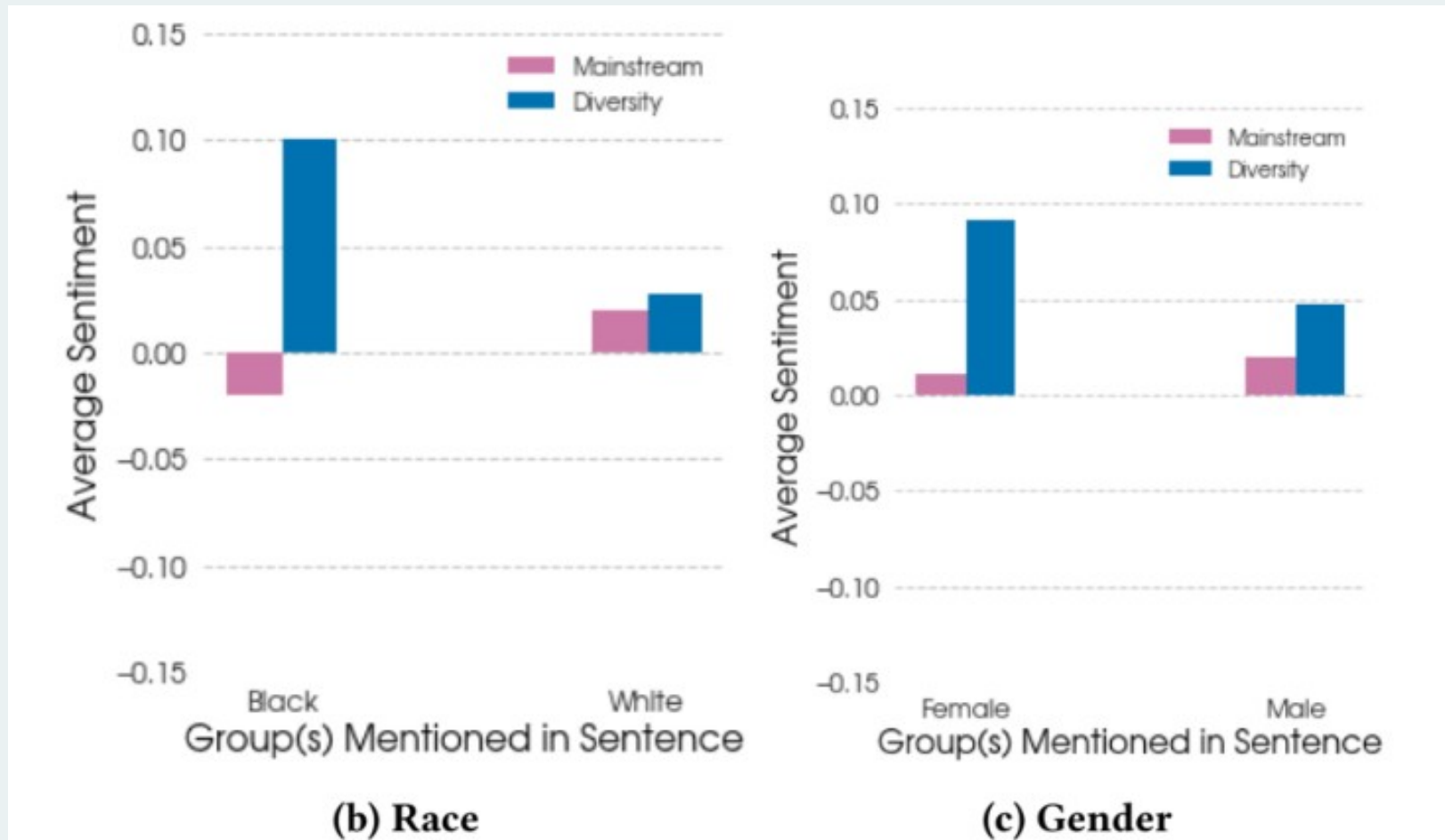
F-Score

- A measure that combines **Precision** and **Recall** into one balanced score
- **Precision** → When the model says “positive,” how often is it right?
- **Recall** → How many of the actual positive cases did it find?

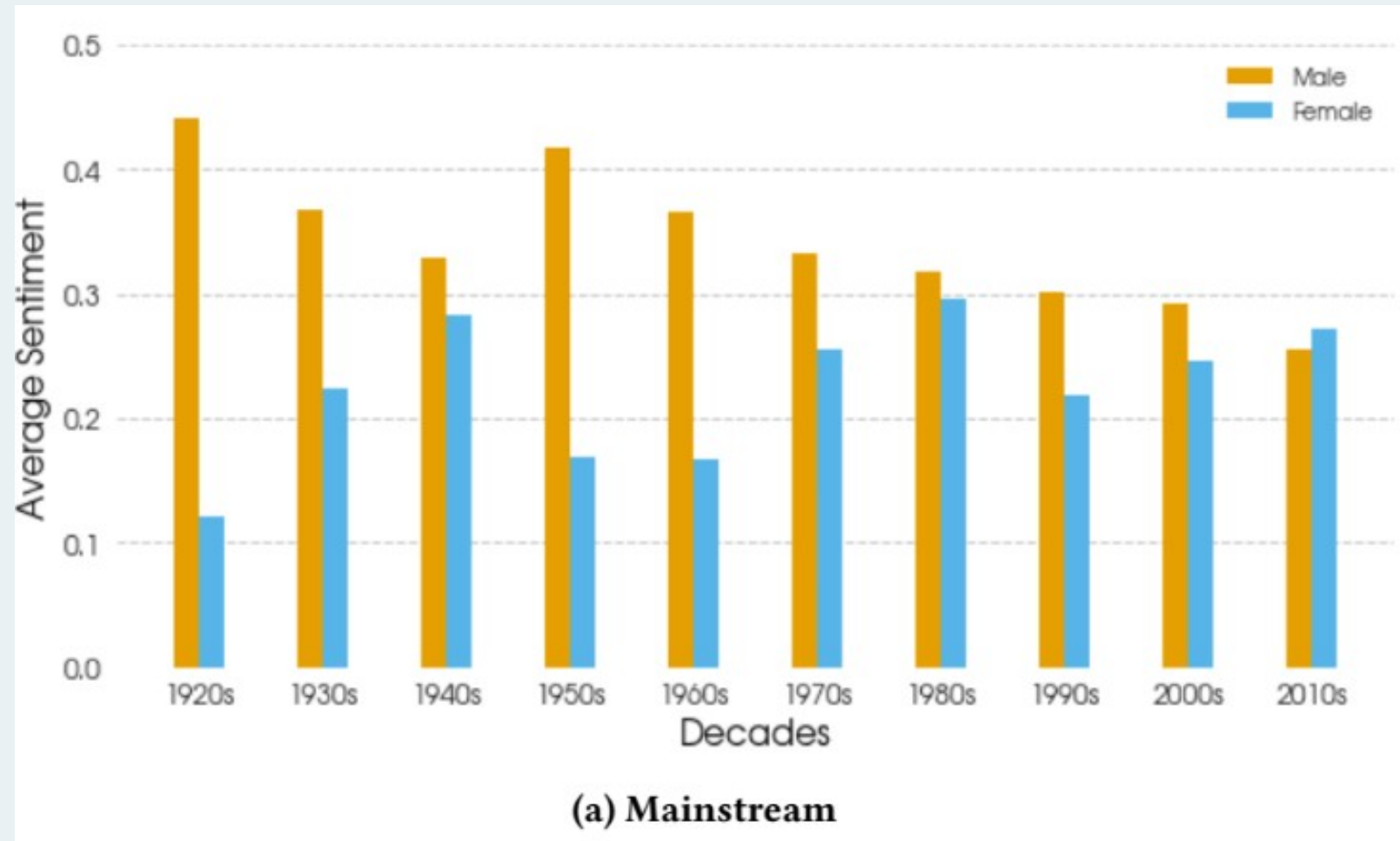
Results



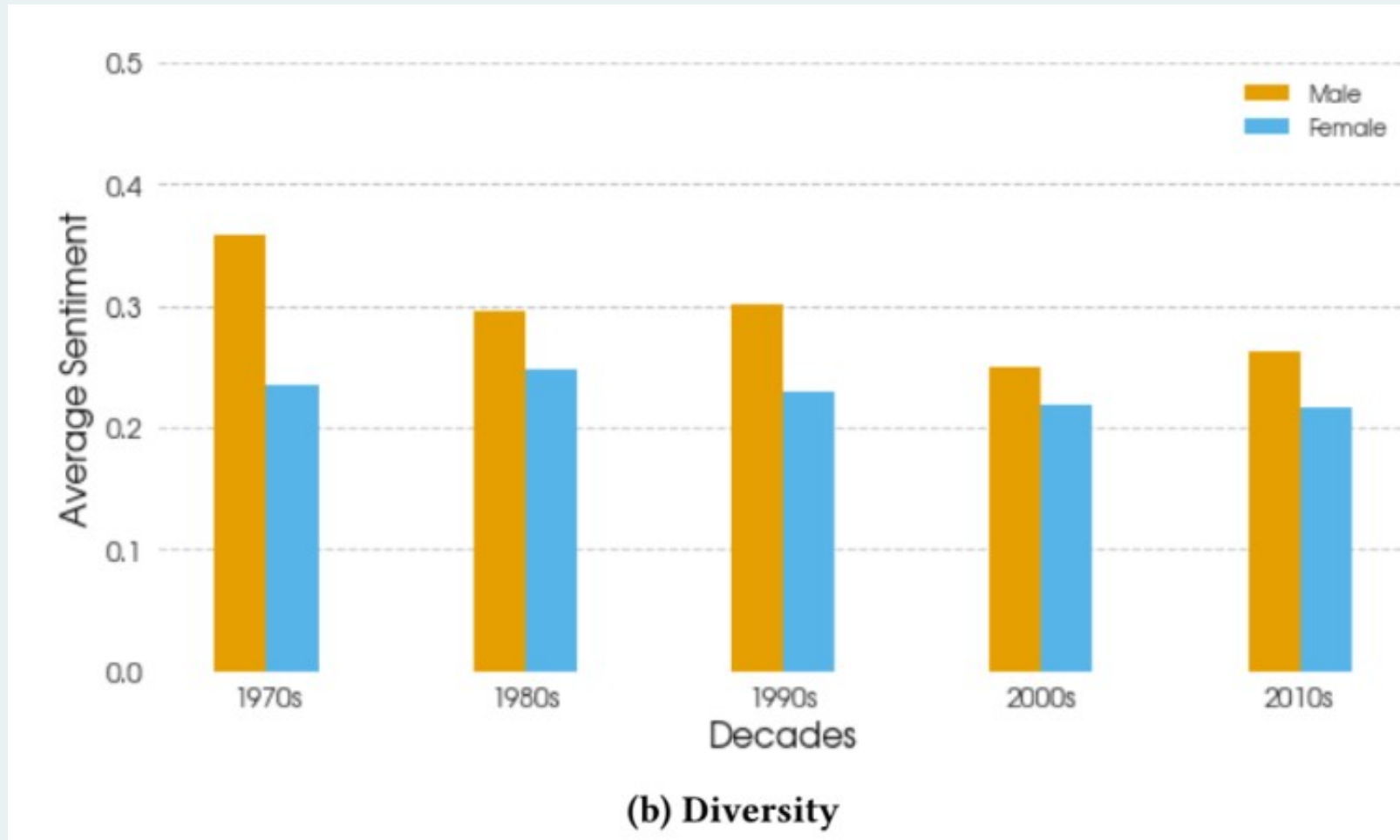
Results



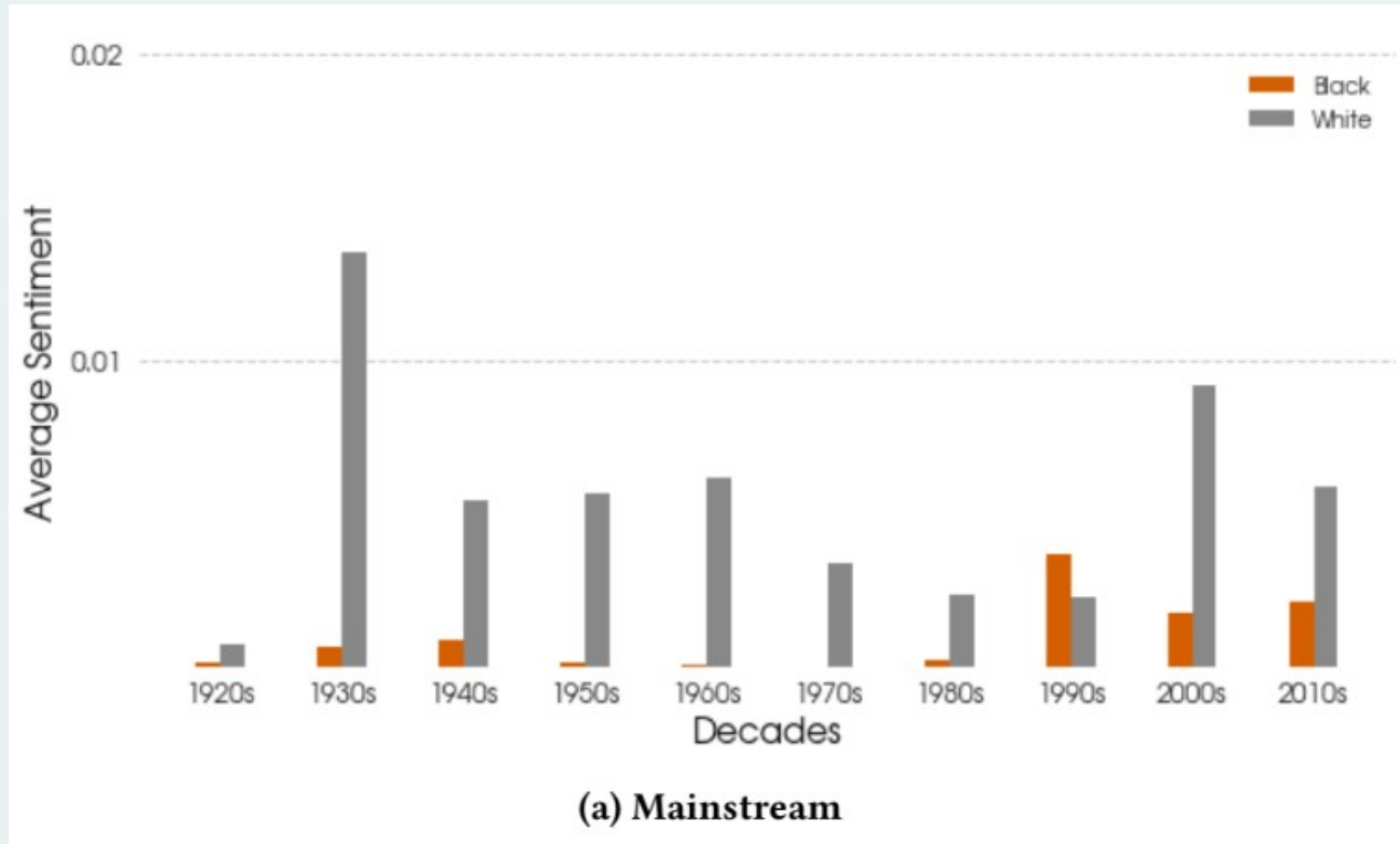
Results



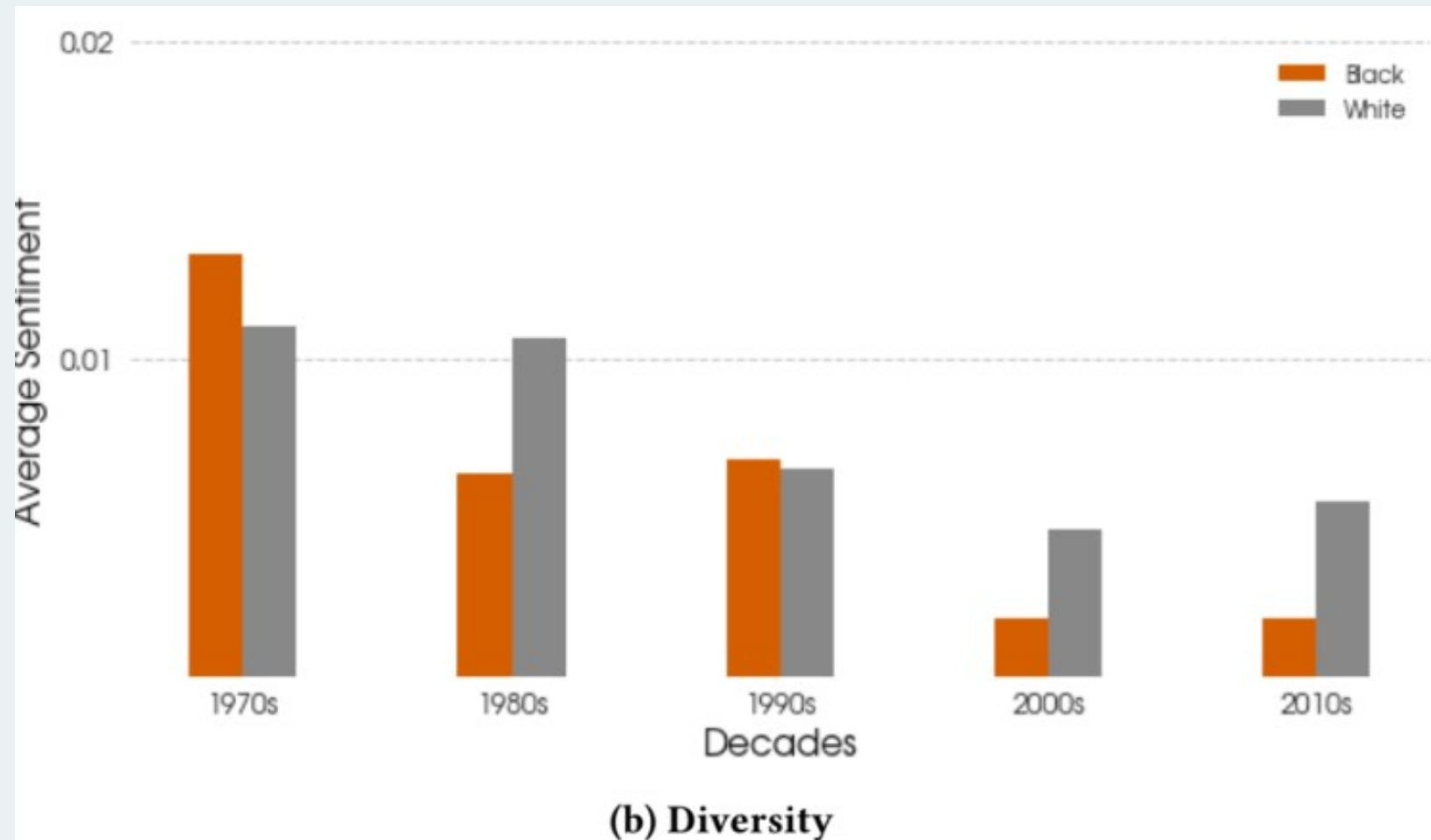
Results



Results



Results



Local Beliefs and Book Consumption

Is there a connection between the way black and white characters are portrayed in children's books and the beliefs of people in the regions where these books are purchased?

- Local beliefs from a large survey (CCES 2017) and compared them with book purchase data (Numerator Omnipanel 2017–2020).

Local Beliefs and Book Consumption

	<i>Dependent variable:</i>	
	% of Respondents who somewhat or strongly agree	
	White people in the U.S. have certain advantages because of the color of their skin	I am angry that racism exists
	(1)	(2)
Sentiment of Black characters	0.131* (0.076)	0.176*** (0.060)
Sentiment of White characters	-0.077 (0.076)	-0.087 (0.060)
Constant	55.871*** (0.293)	80.234*** (0.232)
Observations	17,930	17,930
Adjusted R ²	0.0001	0.0005

Conclusion

- Results are **mostly convincing**
 - effect sizes are small and statistical relationships weak (R^2 near zero)
- Future Research: Studies on **other identities** (sexuality, disability...)
- demonstrates how **machine learning** can help **quantify cultural bias**

Conclusion

- broader implications: highlight the **social responsibility of authors, educators, and publishers**
- broader implications: **audit diversity in teaching materials** and ensure more **inclusive storytelling**
- ethical concerns: risk of **misinterpretation** → “books cause racism”

Discussion



- The paper only included “famous figures” with identifiable race and gender , how might this choice bias the results?
- Do you think that reading diverse children’s books can actually change social attitudes?